

Practical 9

AIM : Association Rule Mining

1. Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
2. Generate association rules from the frequent itemsets and calculate their support and confidence.
3. Interpret and analyze the discovered association rules.

CODE :

```
import pandas as pd
import matplotlib.pyplot as plt

from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

# Create market basket dataset
transactions = [
    ['Milk', 'Bread', 'Butter'],
    ['Bread', 'Butter'],
    ['Milk', 'Bread'],
    ['Milk', 'Bread', 'Eggs'],
    ['Bread', 'Eggs'],
    ['Milk', 'Butter'],
    ['Milk', 'Bread', 'Butter', 'Eggs'],
    ['Bread', 'Butter', 'Eggs'],
    ['Milk', 'Bread', 'Butter'],
    ['Milk', 'Eggs']
]

# Convert transactions into one-hot encoded format
te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)
df = pd.DataFrame(te_array, columns=te.columns_)
print("Transaction Dataset:")
print(df)

# Find frequent itemsets
frequent_itemsets = apriori(
    df,
    min_support=0.3,
    use_colnames=True
)

print("\nFrequent Itemsets:")
print(frequent_itemsets)

# Generate association rules
```

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```
rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.6
)
rules = rules[
    ['antecedents', 'consequents', 'support', 'confidence', 'lift']]
print("\nAssociation Rules:")
print(rules)
# Display rule interpretation
print("\nRule Interpretation:")
for _, rule in rules.iterrows():
    antecedent = ", ".join(rule['antecedents'])
    consequent = ", ".join(rule['consequents'])

    print(
        f"If customers buy [{antecedent}], "
        f"they are likely to buy [{consequent}] "
        f"(Support: {rule['support']:.2f}, "
        f"Confidence: {rule['confidence']:.2f}, "
        f"Lift: {rule['lift']:.2f})"
    )
itemsets = frequent_itemsets.copy()

itemsets["items"] = itemsets["itemsets"].apply(
    lambda x: ", ".join(list(x))
)
plt.figure(figsize=(8, 5))
plt.bar(itemsets["items"], itemsets["support"])
plt.xlabel("Itemsets")
plt.ylabel("Support")
plt.title("Support of Frequent Itemsets")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
rules["rule"] = rules.apply(
    lambda x: ", ".join(x["antecedents"]) + " → " +
    ", ".join(x["consequents"]),
    axis=1
)
plt.figure(figsize=(8, 5))
plt.bar(rules["rule"], rules["confidence"])
plt.xlabel("Association Rules")
plt.ylabel("Confidence")
plt.title("Confidence of Association Rules")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

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OUTPUT :

```
File Edit Selection View Go Run ... AI Practicals
Explorer Problems Output Debug Console Terminal Ports
AI Practicals
  Practical3.py
  Practical4.py
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  Practical7.py
  Practical8.py
  Practical9.py
Outline
Timeline

PS C:\Users\DELL\OneDrive\Desktop\AI Practicals> python Practical9.py
Transaction Dataset:
  Bread  Butter  Eggs  Milk
0   True   True   False  True
1   True   True   False  False
2   True   False  False  True
3   True   False  True   True
4   True   False  True   False
5   False  True   False  True
6   True   True   True   True
7   True   True   True   False
8   True   True   False  True
9   False  False  True   True

Frequent Itemsets:
support      itemsets
0      0.8      frozenset({Bread})
1      0.6      frozenset({Butter})
2      0.5      frozenset({Eggs})
3      0.7      frozenset({Milk})
4      0.5      frozenset({Butter, Bread})
5      0.4      frozenset({Eggs, Bread})
6      0.5      frozenset({Milk, Bread})
7      0.4      frozenset({Butter, Milk})
8      0.3      frozenset({Milk, Eggs})
9      0.3      frozenset({Butter, Milk, Bread})

Association Rules:
  antecedents      consequents      support      confidence      lift
0      frozenset({Butter})      frozenset({Bread})      0.5      0.833333      1.041667
1      frozenset({Bread})      frozenset({Butter})      0.5      0.625000      1.041667
2      frozenset({Eggs})      frozenset({Bread})      0.4      0.800000      1.000000
3      frozenset({Milk})      frozenset({Bread})      0.5      0.714286      0.892857
```

```
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PS C:\Users\DELL\OneDrive\Desktop\AI Practicals> python Practical9.py
8      0.3      frozenset({Milk, Eggs})
9      0.3      frozenset({Butter, Milk, Bread})

Association Rules:
  antecedents      consequents      support      confidence      lift
0      frozenset({Butter})      frozenset({Bread})      0.5      0.833333      1.041667
1      frozenset({Bread})      frozenset({Butter})      0.5      0.625000      1.041667
2      frozenset({Eggs})      frozenset({Bread})      0.4      0.800000      1.000000
3      frozenset({Milk})      frozenset({Bread})      0.5      0.714286      0.892857
4      frozenset({Bread})      frozenset({Milk})      0.5      0.625000      0.892857
5      frozenset({Butter})      frozenset({Milk})      0.4      0.666667      0.952381
6      frozenset({Eggs})      frozenset({Milk})      0.3      0.600000      0.857143
7      frozenset({Butter, Milk})      frozenset({Bread})      0.3      0.750000      0.937500
8      frozenset({Butter, Bread})      frozenset({Milk})      0.3      0.600000      0.857143
9      frozenset({Milk, Bread})      frozenset({Butter})      0.3      0.600000      1.000000

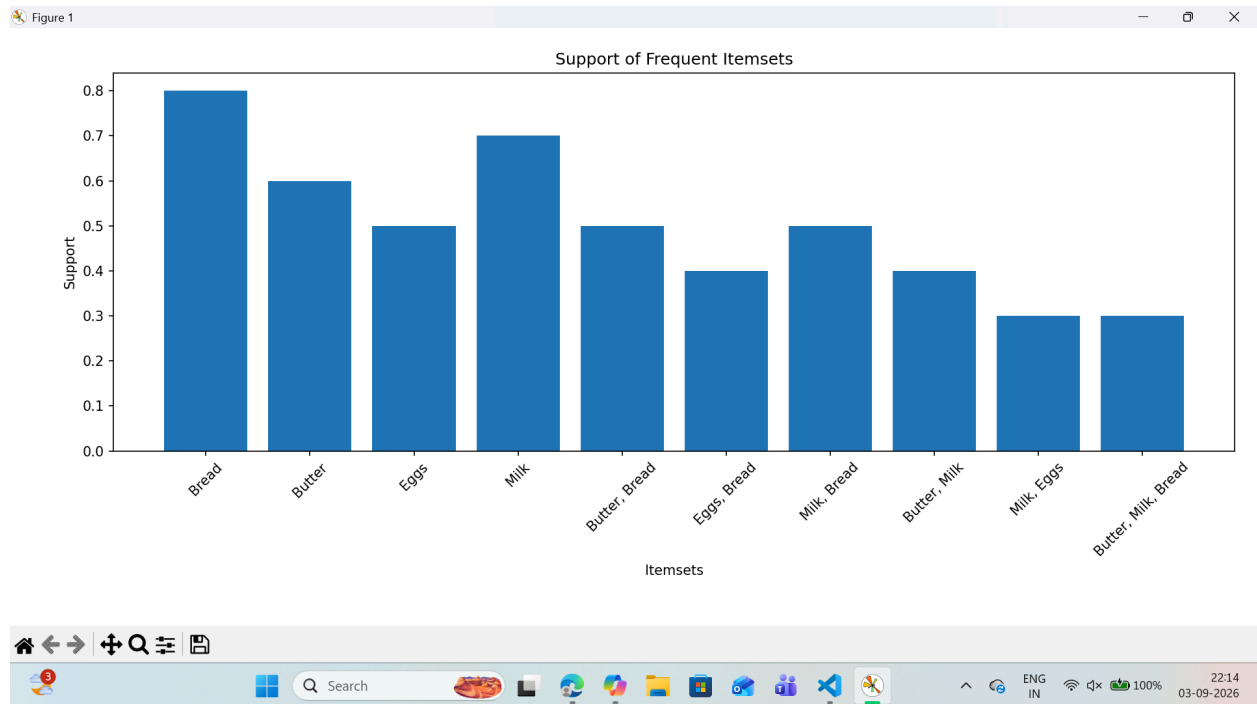
Rule Interpretation:
If customers buy [Butter], they are likely to buy [Bread] (Support: 0.50, Confidence: 0.83, Lift: 1.04)
If customers buy [Bread], they are likely to buy [Butter] (Support: 0.50, Confidence: 0.62, Lift: 1.04)
If customers buy [Eggs], they are likely to buy [Bread] (Support: 0.40, Confidence: 0.80, Lift: 1.00)
If customers buy [Milk], they are likely to buy [Bread] (Support: 0.50, Confidence: 0.71, Lift: 0.89)
If customers buy [Bread], they are likely to buy [Milk] (Support: 0.50, Confidence: 0.62, Lift: 0.89)
If customers buy [Butter], they are likely to buy [Milk] (Support: 0.40, Confidence: 0.67, Lift: 0.95)
If customers buy [Eggs], they are likely to buy [Milk] (Support: 0.30, Confidence: 0.60, Lift: 0.86)
If customers buy [Butter, Milk], they are likely to buy [Bread] (Support: 0.30, Confidence: 0.75, Lift: 0.94)
If customers buy [Butter, Bread], they are likely to buy [Milk] (Support: 0.30, Confidence: 0.60, Lift: 0.86)
If customers buy [Milk, Bread], they are likely to buy [Butter] (Support: 0.30, Confidence: 0.60, Lift: 1.00)
PS C:\Users\DELL\OneDrive\Desktop\AI Practicals>
```

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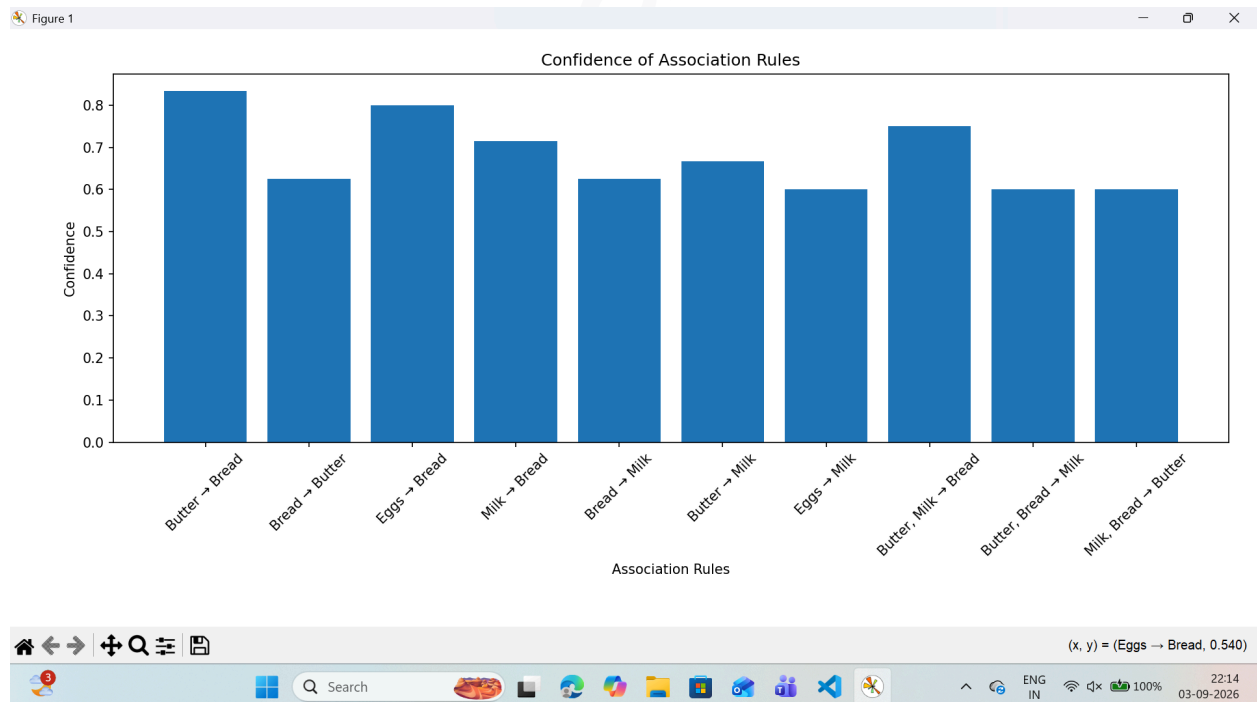
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Support of Frequent Itemsets



Confidence of Association Rules



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